ChIP-seq data analysis handbook

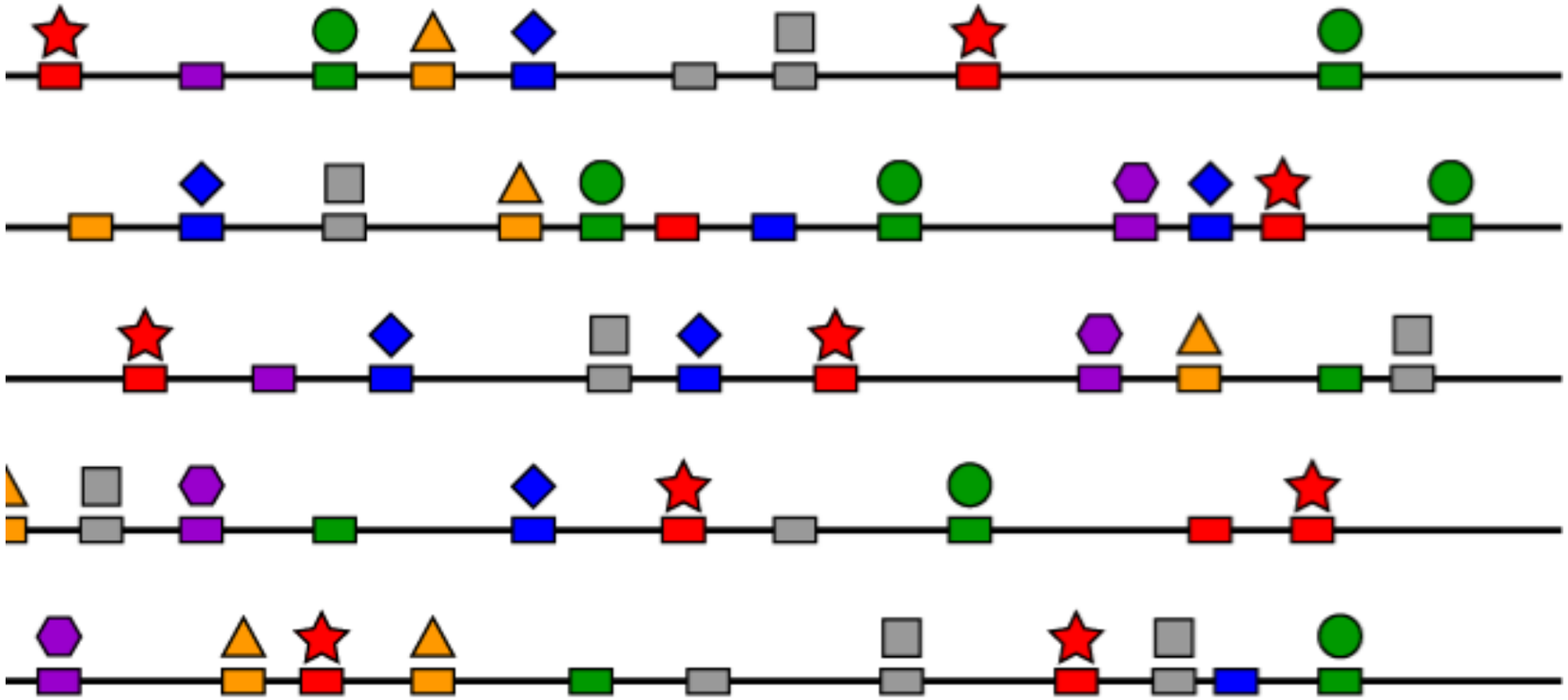
Chengchen Zhao
Shiyang Zeng
2017.12.13

Outline

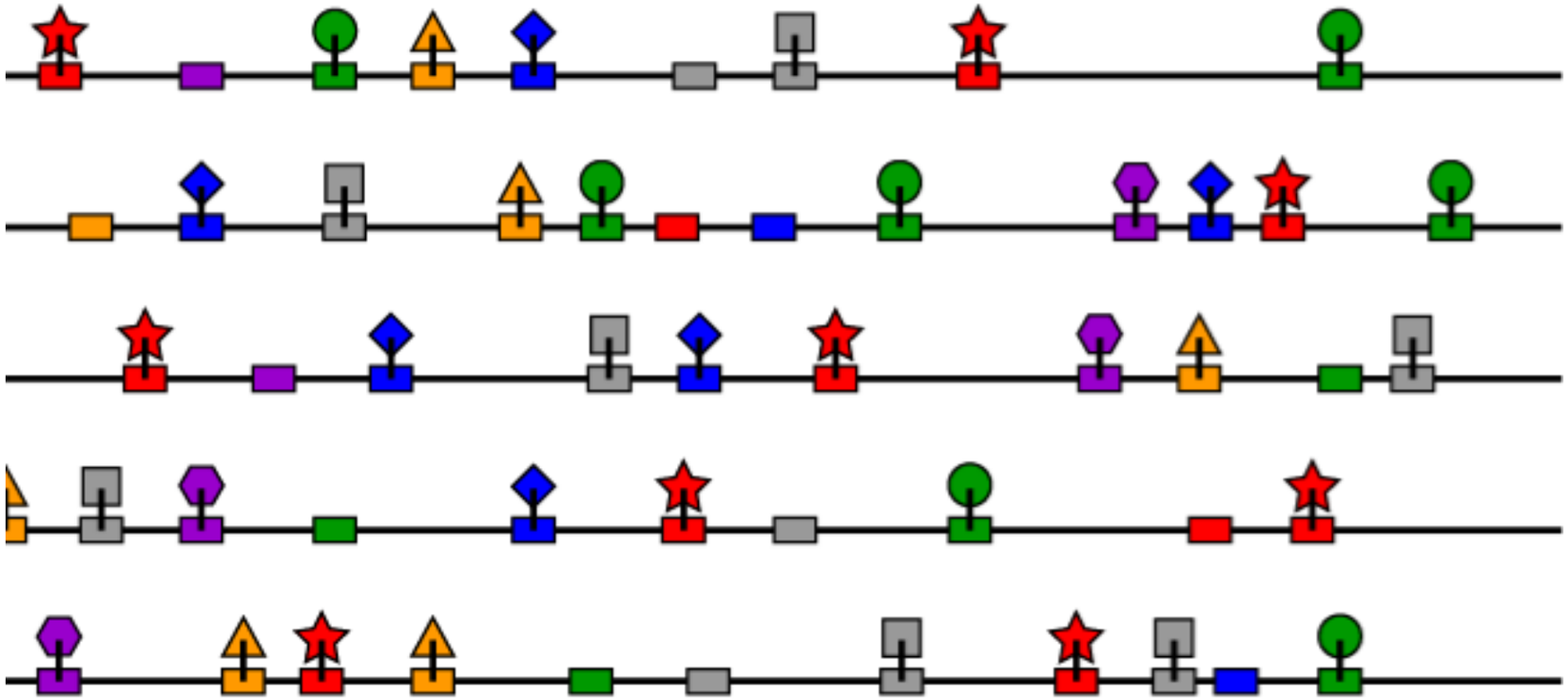
- Introduction
- Analysis workflow
  - Mapping
  - Peak calling
  - Quality control
  - Motif finding
- Downstream optional analysis
  - Beta

Introduction

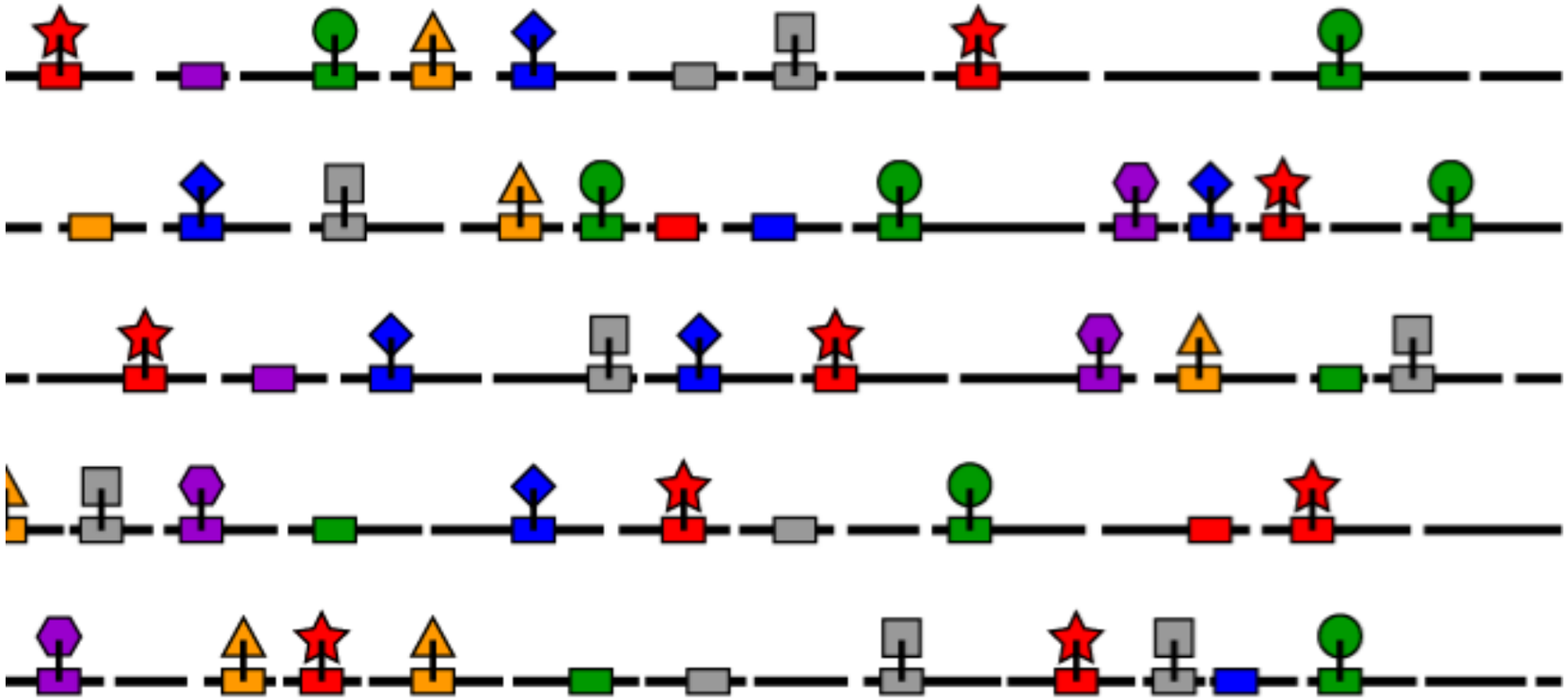
Introduction : the experimental process of ChIP-seq



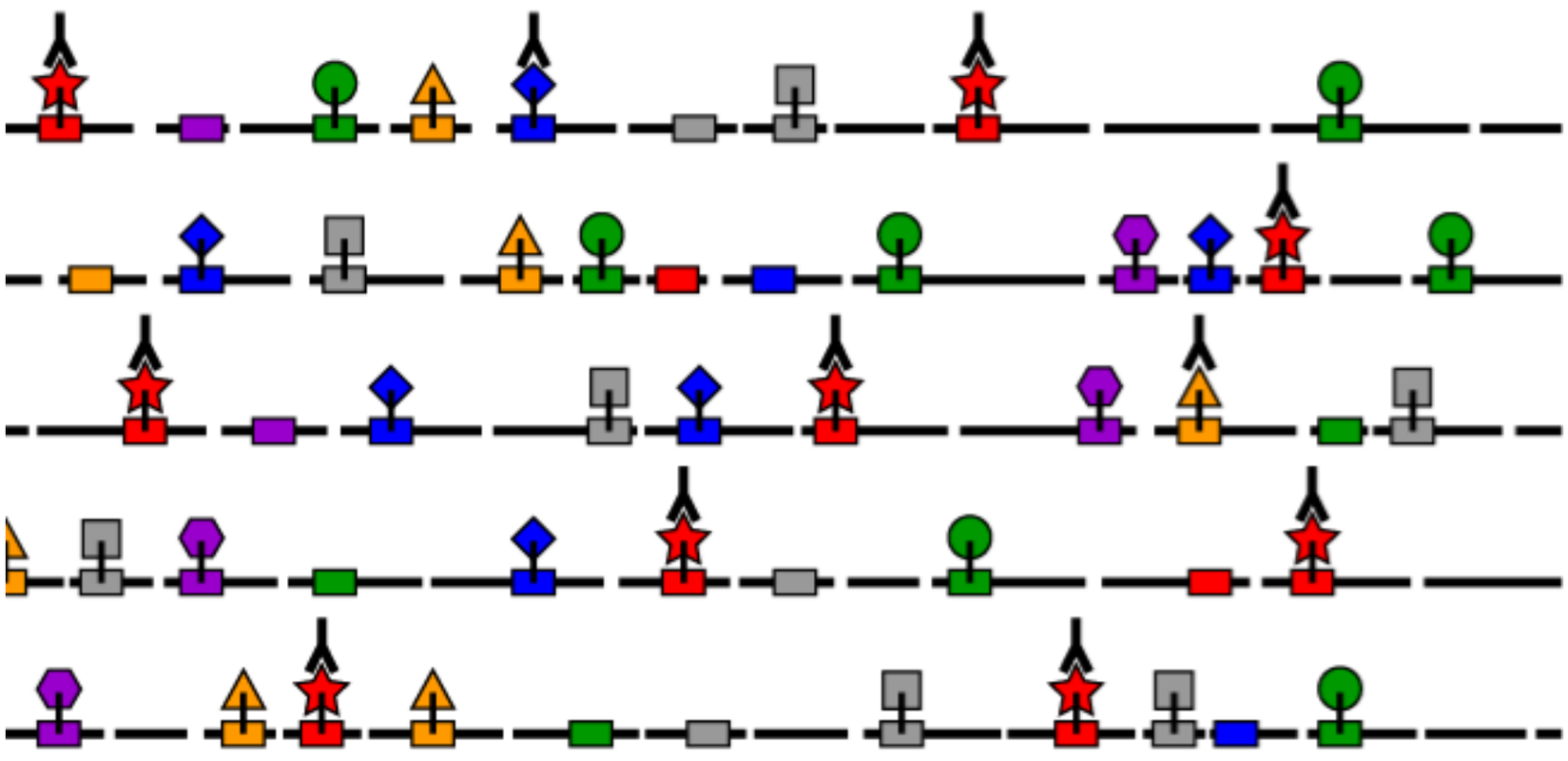
1.TF/DNA Crosslinking in vivo



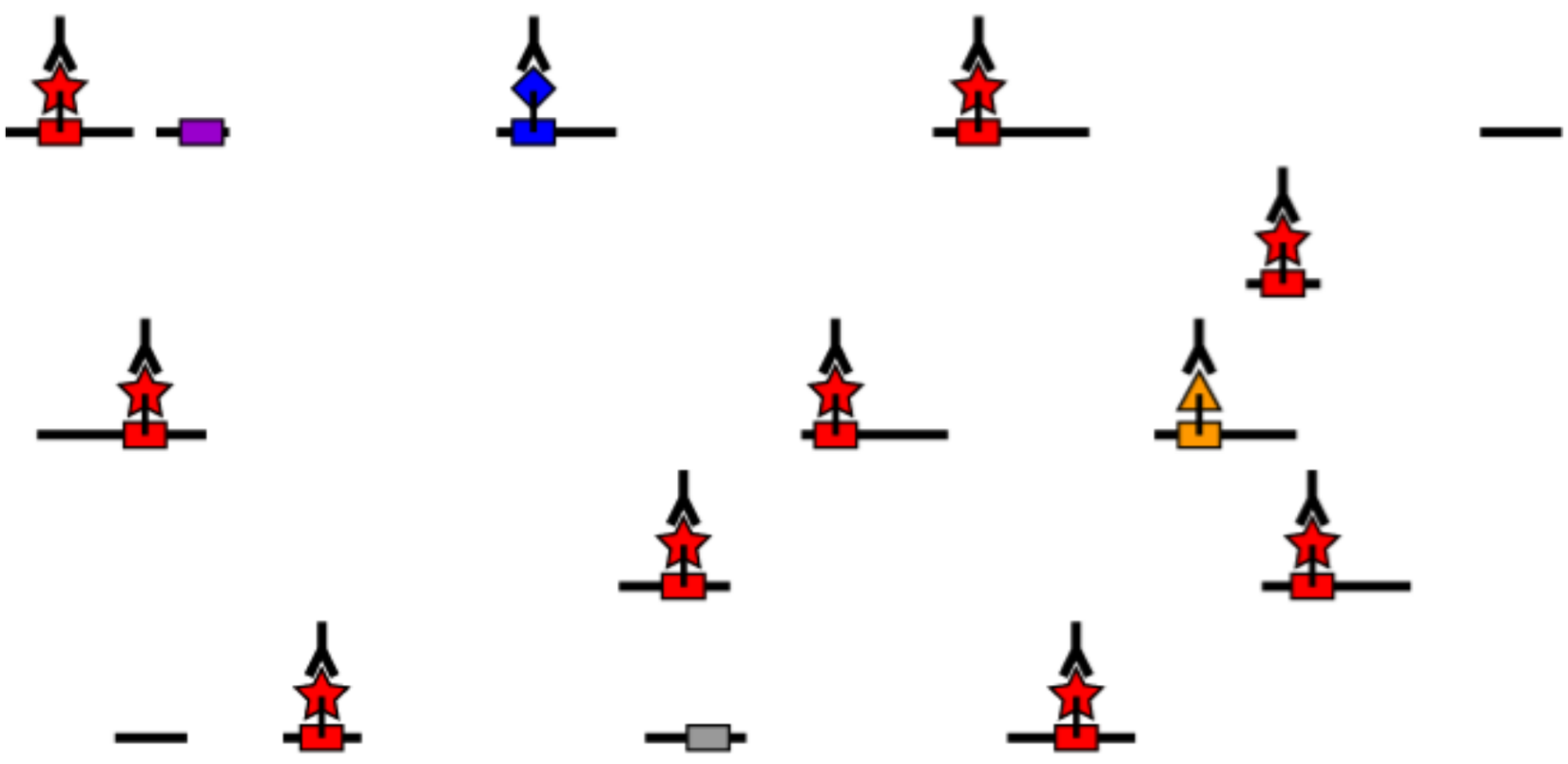
2.Sonication



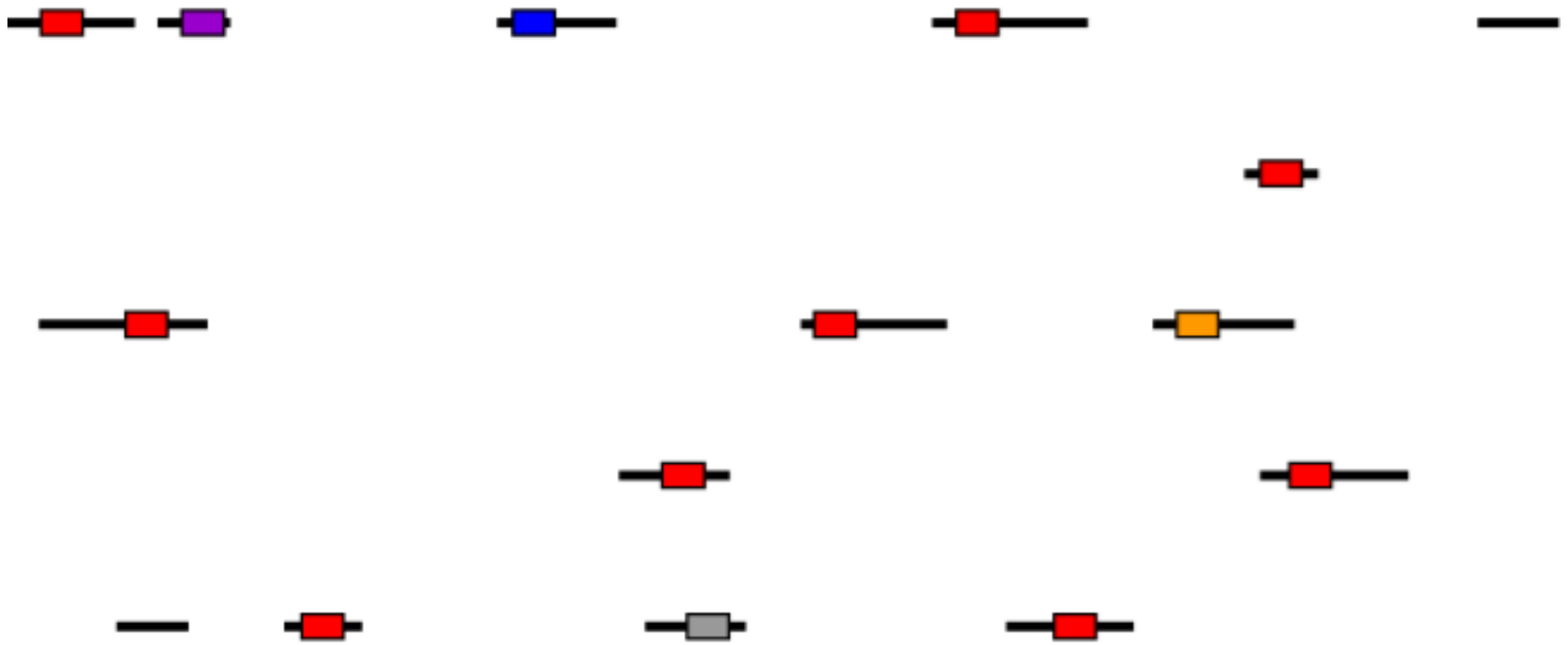
3. Binding with TF-specific Antibody



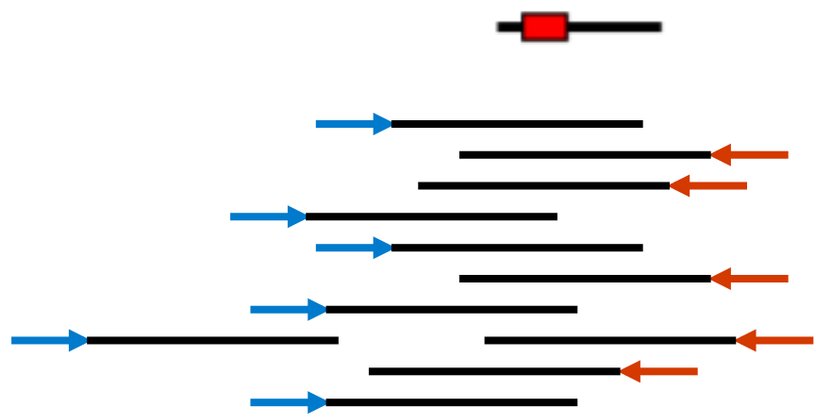
4.Immunoprecipitation



5.Reverse Crosslink and DNA Purification

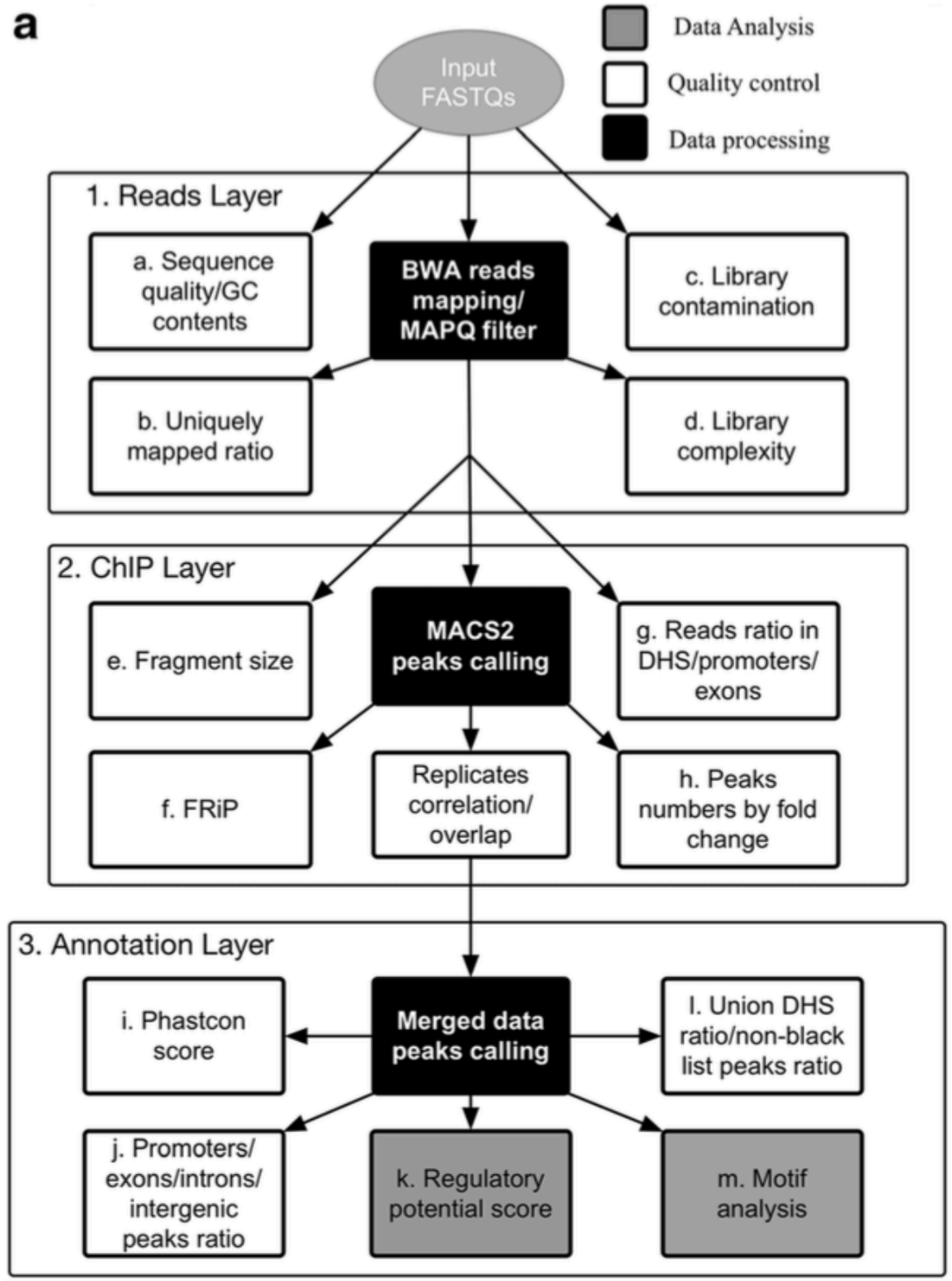


6.Sequencing



Analysis workflow

- Mapping
- Peak calling
- Quality control
- Motif finding



Qian Qin, et.al. 2016

Prior knowledge preparation

Local genomic files needed for analysis

- Mapping index
  - Bowtie2 index: *.bt2
  - BWA index: *.bwt2
- Genome annotation related files
  - Whole genome fasta: *.fa
  - Chromosome length information: *.len
  - Gene models annotation from Refseq: *.gtf, *.refGene

Tools needed for analysis

- samtools
  - Download:
    - http://samtools.sourceforge.net/
  - SAM format:
    - https://samtools.github.io/hts-specs/SAMv1.pdf
- bedtools
  - Download:
    - http://bedtools.readthedocs.io/en/latest/
  - BED format:
    - https://genome.ucsc.edu/FAQ/FAQformat.html

Server common examples of bedtools

intersectBed

```
Chromosome  ================================================================
BED/BAM  A      ====================              =============
BED File B                    ========
Result                        ========
```

closestBed

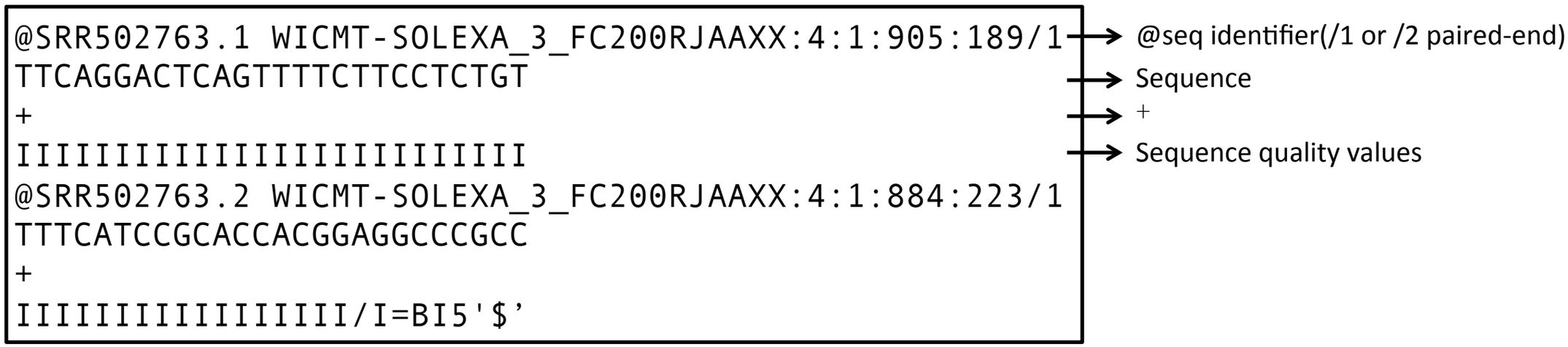
```
Chromosome  ================================================================
BED File A                                  =============
BED File B          ========                                ======
Result                                                      ======
```

coverageBed

```
Chromosome  ================================================================
BED File B  ================     ================     ======    =============
BED File A  ====  ====                 ==                ========    ===  ==  ====
              ========                                              =====  =====  ==


Result      [  N=3, 10/15 ]     [   N=1, 2/16  ]    [N=1,6/6]  [N=5, 11/12 ]
```

Fastq format



Sequence Phred quality score

Phred quality scores Q are defined as a property which is logarithmically related to the base-calling error probabilities P.

$$Q = -10 \times \log_{10} P$$

Phred quality scores are logarithmically linked to error probabilities

Phred Quality Score	Probability of incorrect base call	Base call accuracy
10	1 in 10	90 %
20	1 in 100	99 %
30	1 in 1000	99.9 %
40	1 in 10000	99.99 %
50	1 in 100000	99.999 %

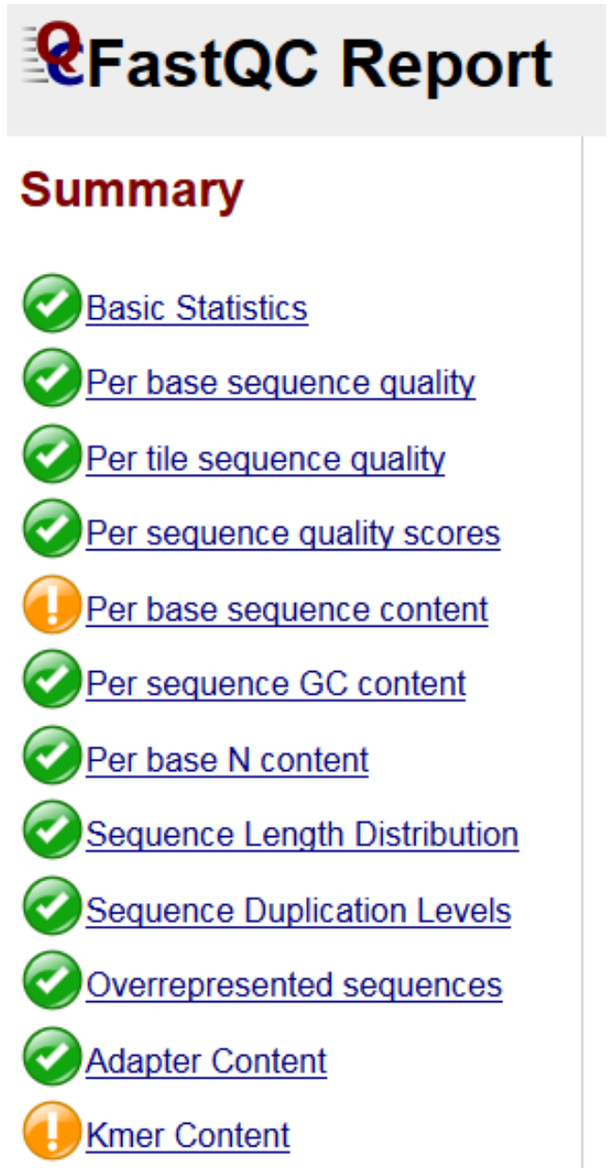
ASCII encoding

S - Sanger Phred+33, raw reads typically (0, 40)
X - Solexa Solexa+64, raw reads typically (-5, 40)
I - Illumina 1.3+ Phred+64, raw reads typically (0, 40)
J - Illumina 1.5+ Phred+64, raw reads typically (3, 40)
L - Illumina 1.8+ Phred+33, raw reads typically (0, 41)

```
 !"#$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~
 |                         |    |        |                              |                     |
33                        59   64       73                            104                   126
```

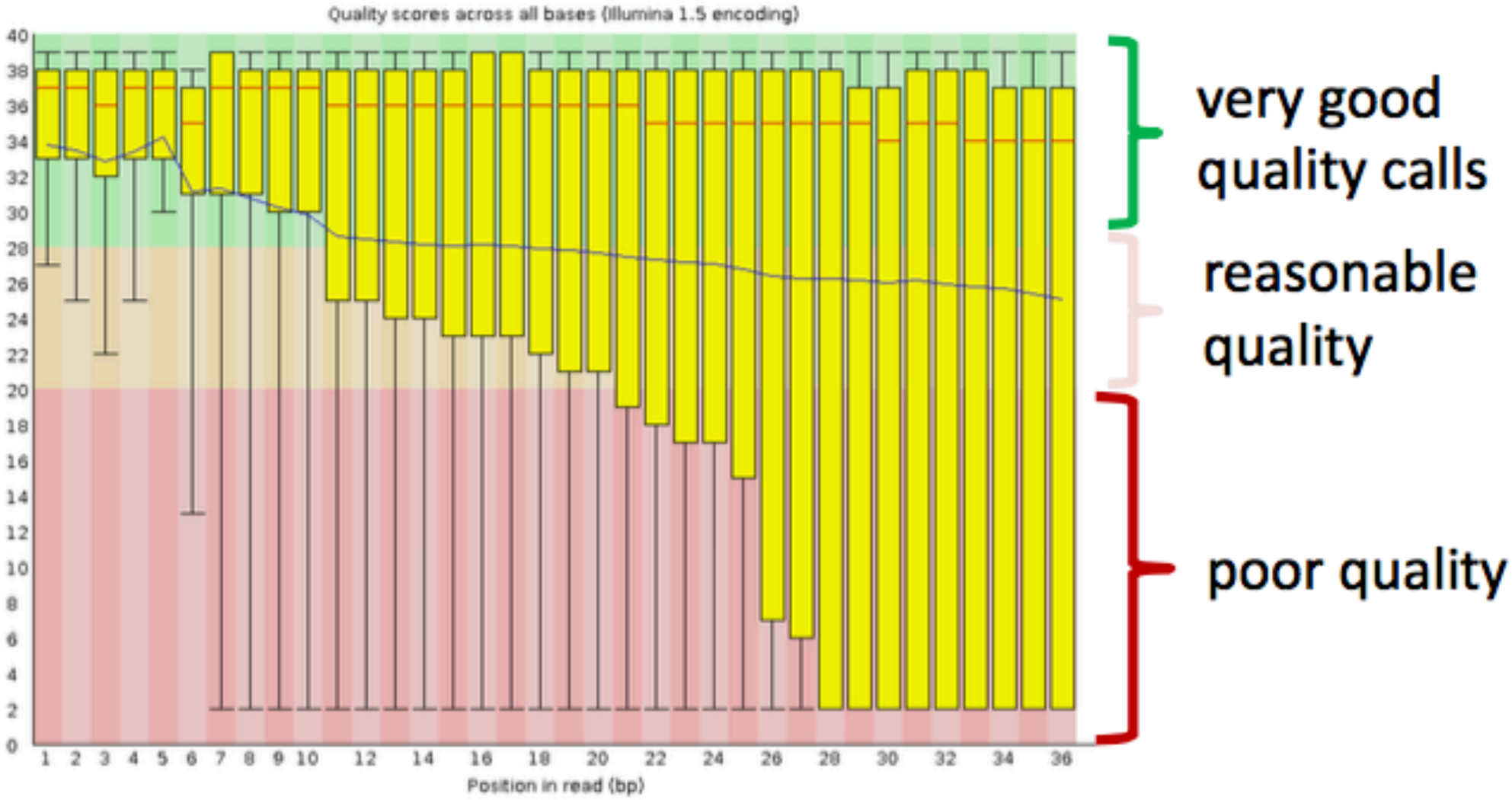
Reads level quality control -- FastQC

- http://www.bioinformatics.babraham.ac.uk/projects/fastqc/
- Run FastQC to check reads quality
  - Input command line:
    - fastqc input.fastq
  - Ouput format: html webpage



FastQC output: per base sequence

- Data example



Quality value = −10 * log10 (error probability)

Quality = 10 => error rate = 10% => base call has 90% confidence

Quality = 20 => error rate = 1% => base call has 99% confidence

Quality = 30 => error rate = 0.1% => base call has 99.9% confidence

FastQC output: …

Mapping Software

- Bowtie:
  - For reads >50 bp Bowtie 2 is generally faster, more sensitive, and uses less memory than Bowtie 1.
  - Bowtie 2 supports gapped alignment, it makes it better for snp calling. Bowtie 1 only finds ungapped alignments.
  - Bowtie 2 supports a "local" alignment mode, in addition to the “end-to-end" alignment mode supported by bowtie1. However we don’t recommend "local" alignment mode for mapping of ChIP-seq data.
- BWA:
  - Bwa is a software package containing several related algorithms using the Burrows-Wheeler Transform. It works well even with indels.

Reads mapping by using Bowtie2

- **Command line usage:**

  bowtie2 [options]* -x <bt2-idx> {-1 <m1> -2 <m2> l -U <r>} [-S <sam>]

- **Typical parameters:**

  <bt2-idx> Index filename prefix (minus trailing .X.bt2).
  <m1> Files with #1 mates, paired with files in <m2>.
  <m2> Files with #2 mates, paired with files in <m1>.
  <r> Files with unpaired reads.
  <sam> File for SAM output (default: stdout)
  -p/--threads <int> number of alignment threads to launch (1)

- **Example:**

GSM307618

```
$ bowtie2_index=/mnt/Storage/home/zhaochengchen/Data/mm9/mm9
$ bowtie2 -q -p 10 -x ${bowtie2_index} -1 ${datapath}/${!sample}.R1.clean.fastq.gz -2 ${datapath}/${!
  sample}.R2.clean.fastq.gz -S ${sample}/${sample}.sam
```

Reads mapping by using BWA

Command line usage:

bwa mem [options] <idxbase> <in1.fq> [in2.fq]

Peak calling

i) Read extension and signal profile generation

strand cross-correlation can be used to calculate fragment length

ii) Peak evaluation

Look for fold enrichment of the sample over input or expected background

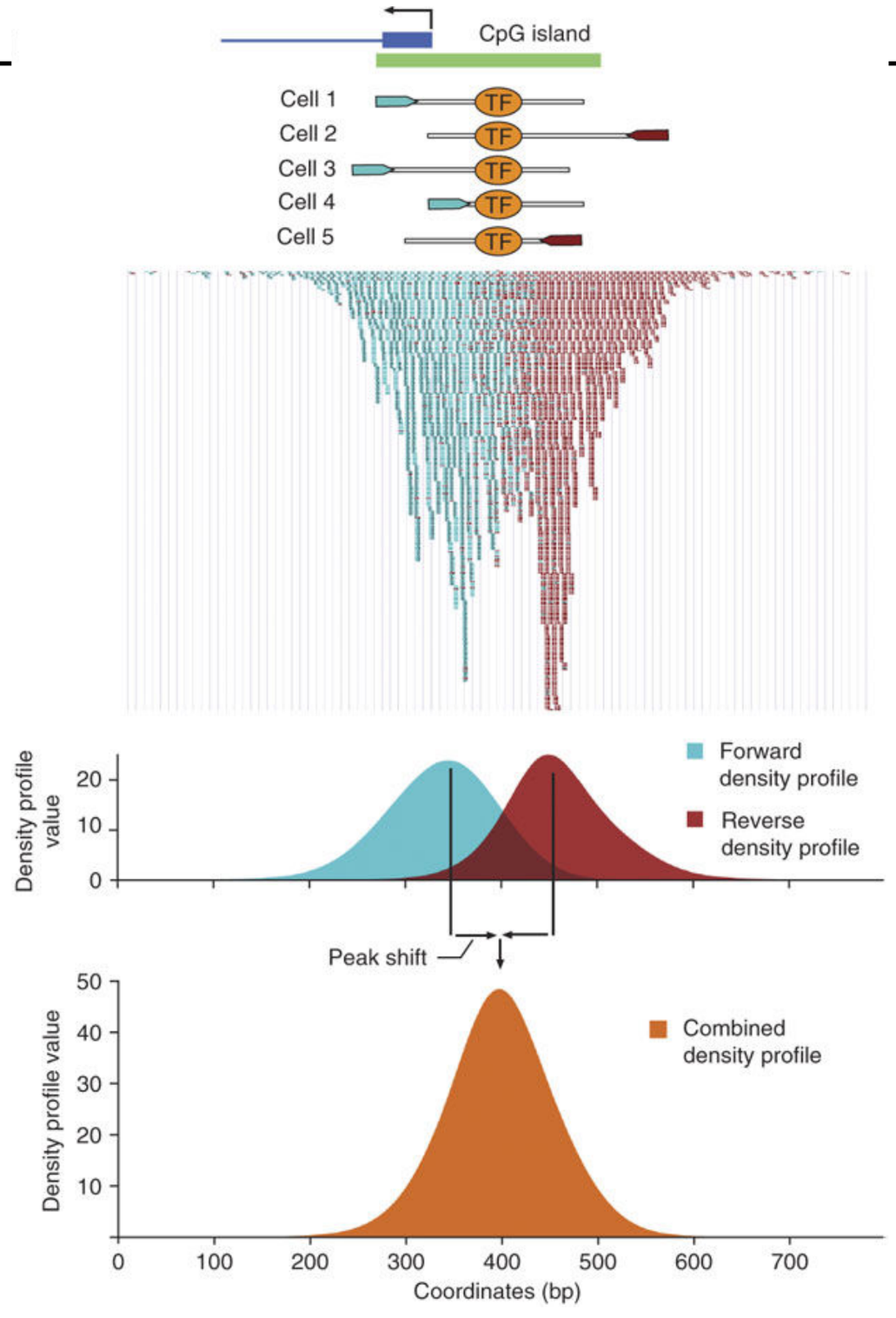
Estimate the significance of the fold enrichment using:

Poisson distribution

negative binomial distribution

background distribution from inputDNA

model background data to adjust forlocal variation (MACS)


CpG island
Cell 1
Cell 2
Cell 3
Cell 4
Cell 5
TF
Forward density profile
Reverse density profile
Density profile value
Peak shift
Combined density profile
Density profile value
Coordinates (bp)

Peak calling by using MACS

- Format converting:

```
$ samtools view -bt ${chrom_info} ${sample}/${sample}.sam -o ${sample}/${sample}.bam
```

- Peak calling using MACS:
  - Histone modification

```
$ macs14 callpeak -g hs -p 1e-5 --keep-dup 1 --nomodel --shiftsize 73 -t ${sample}.bam -c ${control_sample}.bam -n
  ${sample}.macs14
```

  - Transcription factor

```
$ macs14 callpeak -g hs -p 1e-5 --keep-dup 1 -t ${sample}.bam -c ${control_sample}.bam -n ${sample}.macs14
```

Typical parameters for MACS

```
-t TFILE, --treatment=TFILE
                      ChIP-seq treatment files. REQUIRED. When ELANDMULTIPET
                      is selected, you must provide two files separated by
                      comma, e.g.
                      s_1_1_eland_multi.txt,s_1_2_eland_multi.txt
-c CFILE, --control=CFILE
                      Control files. When ELANDMULTIPET is selected, you
                      must provide two files separated by comma, e.g.
                      s_2_1_eland_multi.txt,s_2_2_eland_multi.txt
-n NAME, --name=NAME  Experiment name, which will be used to generate output
                      file names. DEFAULT: "NA"
-f FORMAT, --format=FORMAT
                      Format of tag file, "AUTO", "BED" or "ELAND" or
                      "ELANDMULTI" or "ELANDMULTIPET" or "ELANDEXPORT" or
                      "SAM" or "BAM" or "BOWTIE". The default AUTO option
                      will let MACS decide which format the file is. Please
                      check the definition in 00README file if you choose EL
                      AND/ELANDMULTI/ELANDMULTIPET/ELANDEXPORT/SAM/BAM/BOWTI
                      E. DEFAULT: "AUTO"
-g GSIZE, --gsize=GSIZE
                      Effective genome size. It can be 1.0e+9 or 1000000000,
                      or shortcuts:'hs' for human (2.7e9), 'mm' for mouse
                      (1.87e9), 'ce' for C. elegans (9e7) and 'dm' for
                      fruitfly (1.2e8), Default:hs
--nomodel             Whether or not to build the shifting model. If True,
                      MACS will not build model. by default it means
                      shifting size = 100, try to set shiftsize to change
                      it. DEFAULT: False
--shiftsize=SHIFTSIZE
                      The arbitrary shift size in bp. When nomodel is true,
                      MACS will use this value as 1/2 of fragment size.
                      DEFAULT: 100
```

MACS14 output

Output files:

1. Excel peaks file ("_peaks.xls") contains the following columns

```
Chr,start,end,length,abs_summit,pileup,-LOG10(pvalue),-LOG10(qvalue),name
```

2. "_summits.bed": contains the peak summits locations for every peaks. The 5th column in this file is -log10qvalue

3. "_peaks.narrowPeak" is BED6+4 format file. Contains the peak locations together with peak summit, fold-change, pvalue and qvalue.

To look at the peaks on a genome browser you can upload one of the output bed files or you can also make a bedgraph file with columns :

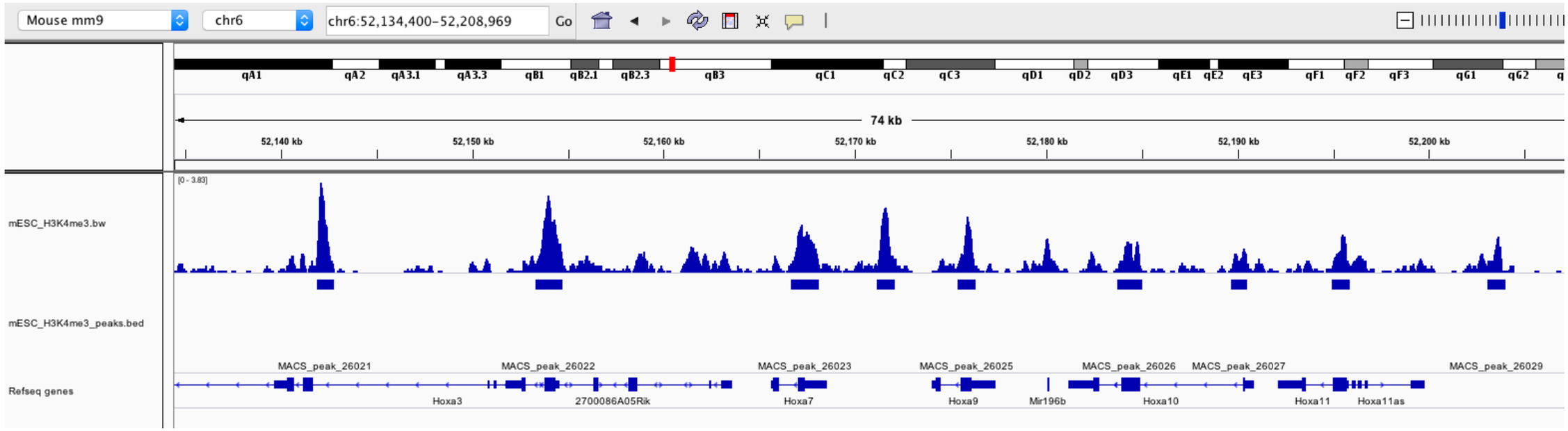
```
chr,  start, end, peak_name, fold_enrichment
```

```
chr1    3141091 3141579 MACS_peak_1     54.27
chr1    3334651 3335282 MACS_peak_2     203.69
chr1    3521726 3522255 MACS_peak_3     116.69
chr1    3660483 3661611 MACS_peak_4     287.17
chr1    3661737 3662795 MACS_peak_5     286.52
chr1    4074970 4075969 MACS_peak_6     243.22
```

Visualisation of ChIP-seq data

- Bigwig file generation

```
$ macs2 callpeak -g hs -B -q 0.05 --keep-dup 1 --nomodel --extsize 147 -t ${sample}/${sample}.bam -n ${sample} --
  outdir ${sample}
$ bedClip ${sample}/${sample}_treat_pileup.bdg ${chrom_info} ${sample}/${sample}.tmp.bdg
$ grep -v _ ${sample}/${sample}.tmp.bdg > ${sample}/${sample}.bdg
$ bedGraphToBigWig ${sample}/${sample}.bdg ${chrom_info} ${sample}/${sample}.bw
```



Normalization of ChIP-seq signal

- MACS can save signal per million reads for fragment pileup profiles with --SPMR.

```
$ macs2 callpeak --SPMR -g hs -B -q 0.05 --keep-dup 1 --nomodel --extsize 147 -t bam/${sample}.bam -n ${sample} --outdir ${sample}.normalise
$ bedClip ${sample}.normalise/${sample}_treat_pileup.bdg ${chrom_info} ${sample}.normalise/${sample}.tmp.bdg
$ grep -v _ ${sample}.normalise/${sample}.tmp.bdg > ${sample}.normalise/${sample}.bdg
```

analysis of cis-regulatory elements

```
$   ceasBW -w normalise.bw/${sample}.bw -b ${sample}.macs14_peaks.bed -g ${refGene} -l ${chrom_info}
```

Motif finding

- Motif finding software
  - MDSeqPos

```
$ for sample in SAMPLE1 SAMPLE2
$ do
$     mkdir ${sample}
$     MDSeqPos.py       -w 600 \
$                       -p 0.001 \
$                       -m cistrome.xml \
$                       -s hg \
$                       -n ${sample} \
$                       -O ${sample} \
$                       ${sample}_summits.bed \
$                       hg19
$ done
```

  - Meme

```
$ fastaFromBed -fi /mnt/Storage/home/zhaochengchen/Data/hg19/hg19.fa -bed ${sample}_peaks.bed -fo ${sample}_peaks.fa
$ meme ${sample}_peaks.fa -dna -oc ${sample} -nostatus -time 7200 -maxsize 1000000 -mod zoops -nmotifs 30 -minw 6 -maxw 50 -revcomp
```

Downstream optional analysis

- Beta

```
$ for sample in SAMPLE1
$ do
$     mkdir ${sample}_BETA
$     BETA minus          -p ${sample}_peaks.bed \
$                               -g hg19 \
$                               -n ${sample} \
$                               -o ./${sample}_BETA
$ done
```

Comparing ChIP-seq across samples i.e. Co-localization or differential binding

- To compare two samples you can use :
- 1. intersectBed (finds the subset of peaks common in 2 samples or unique to one them)
- 2. macs2 bdgdiff (find peaks present only in one of the samples)